

CAPE SPENCER, AK, USA

WMO: 997804

Lat: 58.1989N

Lon: 136.6400W

Elev: 26

StdP: 101.01

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.3	-6.1	-17.7	0.8	-6.5	-14.9	1.0	-3.5	26.3	-4.1	23.6	-4.6	16.7	70	0.923

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.9	16.1	11.6	14.7	11.9	13.9	11.9	13.7	14.7	13.0	14.0	12.6	13.5	5.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	9.6	13.9	12.7	9.2	13.3	12.1	8.8	12.9	38.3	14.7	36.9	14.0	35.6	13.7	16.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.7	16.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	2.1	1.0	2.2	4.5	7.8	10.0	11.4	11.6	10.5	7.2	4.0	2.4
	DBStd	4.44	3.37	3.05	2.95	1.78	1.78	1.65	1.27	1.32	1.52	1.71	2.55	2.88
	HDD10.0	1512	246	253	243	164	72	17	1	1	11	87	181	237
	HDD18.3	4411	504	486	501	413	327	250	215	209	236	344	431	495
	CDD10.0	143	0	0	0	0	4	18	44	51	25	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	5	0	0	0	0	0	1	3	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.8	10.7	9.9	9.3	7.2	5.8	5.4	4.8	5.2	6.7	8.4	10.2	10.2
Precipitation	PrecAvg	1814	188	148	131	92	75	61	85	117	216	259	221	220
	PrecMax	2424	336	237	298	169	156	112	156	218	306	411	408	434
	PrecMin	1474	83	51	21	25	18	23	25	44	77	119	72	115
	PrecStd	226	62	54	56	36	35	20	33	44	48	67	72	87
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	6.6	9.7	10.6	16.1	20.2	17.6	20.0	16.8	11.9	9.4	7.8
		MCWB	7.5	5.0	6.9	6.4	8.4	12.5	11.9	12.5	10.5	10.5	7.9	6.8
	2%	DB	8.1	5.8	7.3	8.6	12.9	14.7	15.1	15.7	14.6	10.9	8.5	7.2
		MCWB	6.7	4.6	5.8	5.7	7.7	11.5	13.3	12.0	10.7	9.7	7.1	6.1
	5%	DB	6.6	5.1	6.1	7.7	11.3	13.4	14.2	14.5	13.5	10.2	7.6	6.6
		MCWB	5.5	4.0	4.5	5.3	8.4	11.3	12.8	12.1	11.3	8.8	6.1	5.4
	10%	DB	5.7	4.3	5.2	7.0	10.1	12.4	13.5	13.7	12.8	9.5	6.9	6.0
		MCWB	4.6	3.1	3.5	4.9	8.1	10.8	12.3	12.1	11.2	8.0	5.6	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	5.7	7.6	7.4	10.9	13.4	14.6	14.6	13.4	11.1	8.6	7.2
		MCDB	9.2	5.9	9.1	10.0	13.4	16.8	15.3	15.9	14.0	11.6	9.0	7.3
	2%	WB	7.1	4.7	6.5	6.6	9.9	12.6	13.9	13.6	12.6	10.2	7.7	6.5
		MCDB	8.1	5.1	7.6	8.3	11.7	15.0	14.5	14.7	13.3	10.8	8.2	7.0
	5%	WB	6.2	3.6	5.5	5.8	9.1	11.9	13.2	12.9	11.9	9.1	6.6	5.7
		MCDB	6.9	4.2	6.8	7.3	10.7	13.5	13.9	13.8	12.8	9.9	7.3	6.4
	10%	WB	5.1	2.9	4.1	5.4	8.7	11.2	12.7	12.4	11.4	8.3	5.9	5.0
		MCDB	5.6	3.5	4.7	6.8	10.1	12.5	13.4	13.3	12.4	9.3	6.7	5.8
Mean Daily Temperature Range	5% DB	MDBR	2.6	2.7	2.7	3.3	3.5	3.7	3.8	3.9	3.4	2.5	2.5	2.3
		MCDBR	2.7	3.5	3.6	4.3	5.6	5.8	5.2	5.5	4.8	2.9	3.1	2.5
		MCWBR	2.3	3.1	2.5	2.8	3.1	3.3	3.8	3.8	3.2	2.7	2.8	2.3
	5% WB	MCDBR	2.7	3.4	3.5	4.0	4.3	4.9	4.7	4.6	3.9	2.8	2.9	2.2
		MCWBR	2.5	3.2	2.5	2.8	2.9	3.2	3.8	3.7	3.1	3.0	2.9	2.4
Clear-Sky Solar Irradiance	taub	0.298	0.298	0.323	0.376	0.397	0.383	0.379	0.365	0.345	0.323	0.292	0.261	
	taud	2.165	2.385	2.343	2.217	2.236	2.355	2.420	2.474	2.499	2.427	2.225	2.136	
	Ebn at Noon	519	750	825	827	834	854	850	838	800	705	529	445	
	Edh at Noon	51	65	89	119	125	113	104	92	77	62	47	37	
All-Sky Solar Radiation	RadAvg	0.40	1.16	2.43	3.91	5.03	5.33	4.65	3.64	2.38	1.27	0.53	0.26	
	RadStd	0.05	0.17	0.25	0.31	0.45	0.35	0.34	0.42	0.35	0.17	0.07	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	+1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page